

Quantifying Flood Impacts on Land Use and Land Cover and Predicting Future Landscape Dynamics Using Multi-Temporal Remote Sensing Data and the CA–Markov Model: A Case Study of Khagaria District, Bihar, India

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Abstract

The spatiotemporal analysis of land use and land cover is carried out from 2015 to 2025 using remotely sensed data, including Landsat 8 (2015), Sentinel-2 (2020) and Sentinel-2 (2025) of Khagaria district, Bihar. The study used a random forest classifier to perform supervised classification on the LU/LC map, producing five classes: forest, water bodies, built-up cropland, and barren landscape. The overall accuracy for the LU/LC maps in 2015, 2020, and 2025 was 95%, 91%, and 95%, respectively. It was discovered that the flood was the primary cause of the LU/LC shift. In this area, flooding is a common occurrence. The sector most affected by floods was agriculture, which had a 3.87% decrease after the flood event even though it remained the most agriculturally dominated sector. The built-up area has grown by 5.19%, with some parts farther away and others near flood-prone areas. The Forest/Plantation has shrunk by 1.53%. Floods have resulted in a 0.99% percent decrease in water bodies and a 1.19% percent increase in barren land. The government should enact appropriate legislation, including mandatory insurance coverage for assets located in high-risk areas, to ensure compensation for losses caused by natural disasters. Furthermore, Land Use/Land Cover (LU/LC) analysis plays a critical role in identifying the underlying drivers of adverse land cover change, thereby supporting evidence-based policymaking and sustainable land management.

Keywords: Land use land cover, LU/LC Change, Flood, Spatio-temporal Change, Geographical Information System (GIS).

Introduction

Land serves as a fundamental material basis for human life and development, being a crucial production element and source of subsistence⁷. It illustrates human endeavours and their effects on the environment²¹. It is a measurable area of the earth's terrestrial surface, encompassing land and its resources, and is delineated by UNEP and FAO as all attributes of the biosphere directly above or below the surface. This includes characteristics of the near-surface environment in habitable regions, such as landscape, soil, surface hydrology (including marshes, swamps, rivers, and shallow lakes), groundwater, and geo-hydrology⁴². The word "land use" denotes human activities and various applications of land, including urban development, agricultural zones, recreational spaces, and wildlife control areas, among others. A fundamental property of the Earth system is land cover, defined as the assemblage of biotic and abiotic components on the planet's surface²⁴. In other words, land use (LU) denotes the human utilization of land and encompasses the strategic planning and management of land as determined by human activities⁶. Conversely, land cover (LC) refers to the natural surface covering of the land as determined by natural processes. It encompasses the distribution of flora, hydrology, arid regions, glacial formations, and the adjacent subsurface, comprising biota, soil, topography, surface, and groundwater²⁴.

A primary issue concerning the evolving global environment is the alteration in land use and land cover (LULC)⁸. Land use and cover change (LUCC) visually represents the effects of human activities and natural conditions on the terrestrial surface, serving as a nexus between human socioeconomic activities and the natural environment, closely associated with global climate change, the surface carbon cycle, and biodiversity⁴². Changes in land use and land cover (LU/LC) have been recognized as a major driver of environmental change across all geographical and temporal scales⁴⁰. Land use and land cover (LULC) change is a dynamic process influenced by natural events and human actions, which subsequently induce alterations affecting natural ecosystems³⁷. Land use delineates human interactions with the land, whereas terrain cover provides overarching details regarding the landscape. For example, while a forest constitutes a land cover category, its classification is dictated by land use. The primary disincentive for residents living near a river is the potential for floods³¹. LULC analysis often seeks to find alterations within the same geographical area throughout two distinct times³⁰. Satellite image-based land cover classification algorithms enable the comparison of images from several timeframes to highlight changes in developed and undeveloped area¹⁷. The CA–Markov model is a hybrid of the CA and Markov models. They are suitable for LULC change detection and predictions⁵ because they take into consideration the spatial and temporal components of land-cover dynamics¹⁶.

Ahmed et al. investigated the immediate and long-term effects of the 2014 flood on Wular Lake utilizing satellite imagery and supervised GIS classification. Terrestrial vegetation (23.7%), agricultural land (43.7%), and aquatic bodies (39.9%) had a substantial decline in land use and land cover images from 25 August 2014 (pre-flood) to 13 September 2015 (post-flood). Superimposing pre- and post-flood imagery with satellite data from September 10, 2014, indicated an area of 88.77 km² that was submerged by flooding. Prior to the inundation, 34.06 km² of aquatic vegetation, 13.89 km² of agricultural land, and 3.13 km² of terrestrial vegetation were submerged. Following the flood, siltation transformed over half of the area beneath water bodies into sand deposits (22.76 km²). Aquatic flora, agricultural land, and sand deposits were observed in flood-inundated images following overlay analysis. Pre- and post-classified images

were utilized to evaluate and quantify alterations in the flood-affected region (88.77 km²). The alterations in land use and land cover indicate that Wular Lake's biodiversity has been adversely affected by flooding³.

The Melamchi-Indrawati river basin, Sidhupalchowk's second-most flood-prone area, was assessed following the flood on June 15, 2021, by Thapa and Prasai. They assessed the effects of the research region by utilizing pre-, post-, and contemporary Land Use and Land Cover (LULC) maps, and analyzed LULC alterations through Google Earth and Sentinel 2 data from February 1 to March 1, 2019; July 1 to August 1, 2021; and February 1 to March 1, 2022. A random forest classifier was employed for supervised categorization into forest, agricultural land, barren terrain, aquatic bodies, and urban areas in our land use and land cover map. The 2019 Land Use Land Cover (LULC) map exhibited an accuracy of 94.26%, whilst the maps for 2021 and 2022 demonstrated accuracies of 92.18% and 92.72%, respectively. Floods affected 90.48 percent of agricultural land in the study area. Deserts have increased by 78.58%. The built-up area decreased by 70.61 percent. Flooding diminished forest coverage by 93.06%. To mitigate expenditures associated with natural disasters, the government ought to provide high-risk land insurance³⁹.

The integration of the Markov Chain model with Cellular Automata provides a robust method for predicting and simulating spatial-temporal land use and land cover changes¹¹. This integrated technique is excellent in predicting and forecasting complicated land use and land cover categories. The CA-Markov hybrid model may proficiently integrate remotely sensed data and Geographic Information Systems (GIS). This approach may transform the outcomes into spatially explicit data essential for land use and land cover development¹². This model has been effectively used in LULC prediction in numerous research. This study looked at and forecasted the flood effects around BN and WN in Khartoum for future floods. Land-use/land-cover (LULC) changes following floods in 2014, 2016, and 2020 were projected by RS and GIS in 2024. We calculated the risk of flooding. BN experienced water, built-up, and agricultural changes of 0.30%, 0.25%, and 0.05% between 2014 and 2016, and 3.38%, 3.68%, and 7.28% between 2016 and 2020. Between 2014 and 2016, the agricultural, built-up, and water changes

around WN were 5.56%, 3.26%, and 8.02%; between 2016 and 2020, they were 13.79%, 5.62%, and 21.40%. According to change evaluation, the area near WN saw the greatest flood damage, possibly encompassing Khartoum. Waterbodies may decrease by 4.18% (BN) and 20.35% (WN) in 2020–2024, according to RS–GIS, despite hydraulic analysis indicating Khartoum's flood risk is significant. Therefore, population dispersion and physical dangers should be taken into consideration in assessments of Khartoum flood risk¹.

Nevertheless, flooding has emerged as a substantial menace, affecting vast areas and communities worldwide. More than 75% of global natural risks are attributed to flooding, frequently yielding immeasurable consequences and affecting 170 million people annually¹⁸. Floods impede the relationship between individuals and the land, affecting land use and land cover change (LULC)²⁶. Khagaria's flood history has been catastrophic, with around 74% of the district's territory susceptible to flooding⁹. Significant floods transpired in 1954, 1957, 1974, 1987, 2004, 2007, and 2017, profoundly affecting individuals' lives, properties, and the agricultural sector of the district. Floodwaters frequently inundate agricultural area, resulting in extensive damage to existing crops. Annually, roughly one hundred thousand hectares of standing crops are obliterated by floods, resulting in substantial losses, including fatalities¹⁹. The LU/LC change analysis would facilitate the enactment of appropriate legislation, such as mandates for insurance on high-risk properties, which the government should enforce to mitigate damages from natural catastrophes. Floods induce significant alterations in the land use/land cover (LU/LC) pattern, and given that Khagaria district experiences annual flooding, these changes necessitate examination for improved LU/LC planning and management. This study seeks to assess the influence of flooding on land use/land cover changes across three years and three decades in Khagaria district of Bihar and predict post-flood LULC changes in 2035. This study will be amongst the first to assess future post-flood LULC changes and simulating land-use changes under different scenarios in the Khagaria district using the cellular automata (CA)–Markov model.

Study Area

Khagaria and Gogari are the two subdivisions that make up the Khagaria district. The district has seven blocks: Alauli, Beldaur, Chautham, Gogari, Khagaria, Parbatta, and Mansi. The coordinates of Khagaria are 25°30'N, 86°29'E to 25.5°N, 86.48°E, and its total area is 1,485 square kilometers. According to the 2011 census, the population of Khagaria was 1,666,886, with 883,786 men and 783,100 women.

The district's major rivers are the Ganga, Burhi Gandak, Bagmati, Kamla, and Ghaghri (the mainstream of the Kosi river). The climate of Khagaria can be described as a transitional zone between the humid atmosphere of Bengal's southern valley and the dry, scorching heat of the northern uplands³². The district has no hills or mineral deposits, and agriculture remains the primary industry. Wheat is the principal crop grown here, reflecting the region's land-use pattern. Figure (1) represents the study area map of Khagaria, which is created using SRTM DEM data in Arc-GIS software. The area has an elevation ranging between 0–91 meters

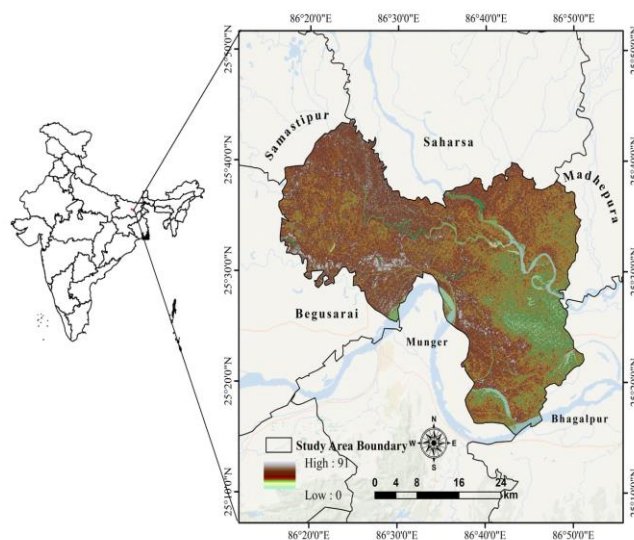


Figure 1. Study Area

Material

Table 1. contains all the data used in this study. The study uses multispectral, multi-temporal Sentinel-2 satellite data and Landsat 8 (Operational Land Imager & TIRS) (30 m) for Khagaria district that were collected for three years, namely, 2014, 2019, and 2024, which was utilised for visual interpretation of land use and land cover via QGIS software. The

Sentinel datasets were acquired via the Copernicus Open Access Hub website (<https://www.copernicus.eu/en/accessdata/conventional-data-access-hubs>). Landsat data was collected from USGS website. OSM Toposheet No.- 72K/11, 72K/10, 72K/6, 72K/7, 72K/14, 72K/15 (2010) with Scale 1:50,000 Satellite data from Survey of India (SOI) has been obtained. The toposheet map was used to locate the actual world feature on the LU/LC map. The field visit in July 2024 gathered all Ground Control Points (GCPs) using Survey of India (SoI, 1:50,000 scale) topographic maps and Google Earth.

Table. 1 Data Type and Sources

Sr. No.	Data Type	Source
	Landsat 8 (Operational Land Imager & TIRS) (30 m spatial resolution) (2015)	USGS Earth Explorer
2.	Sentinel - 2 (10 m spatial resolution) (2020,2025)	Copernicus Open Access Hub
3.	Sentinel - 1 (10 m spatial resolution)	Copernicus Open Access Hub
4.	OSM Toposheet No.- 72K/11, 72K/10, 72K/6, 72K/7, 72K/14, 72K/15 (2010)	Survey of India (SOI)
5.	Google Earth	https://earth.google.com/web/

Methodology

Figure 2 is the flowchart of methodology, which shows the methods used in this study: data acquisition, image preprocessing and classification, accuracy assessment, and LULC map creation, change detection, and future prediction.

Data Acquisition

Multispectral and multitemporal satellite datasets were acquired for Khagaria district for the years 2015, 2020, and 2025. Landsat-8 imagery for 2015 was obtained from the USGS portal, while Sentinel-1 and Sentinel-2 data were downloaded from the Copernicus Open Access Hub. Flood inundation layers were generated using Sentinel-1 datasets corresponding to heavy rainfall events, and these

event-based layers were subsequently integrated to produce an annual composite flood inundation map.

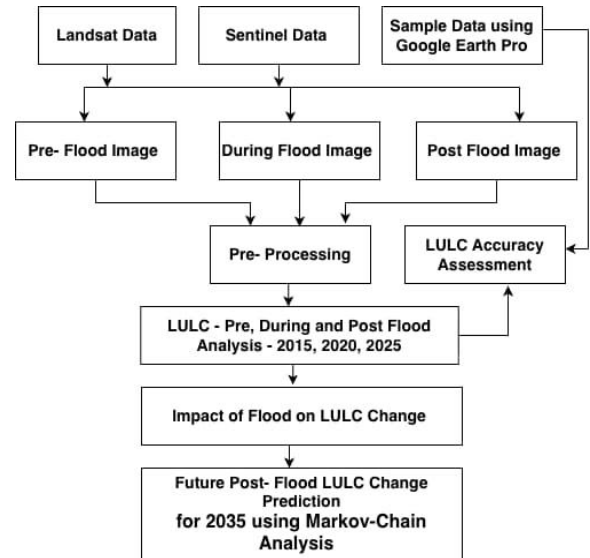


Figure. 2 Flowchart of Methodology

Data Acquisition

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Image Pre-processing

It is essential to preprocess satellite photos before detecting changes to establish a more direct connection between the collected data and biophysical processes². The acquired data has been used to process the LU/LC map of the study area. To understand the topography, water flows, and other physical features like major highways, railway lines, location, and levels on benchmarks, we need the Survey of India topographical maps (1:50,000) of the planning region. The data collected has been georeferenced using WGS 1984, Universal Traverse Mercator (UTM) projection system, and then using ERDAS, all the topographic sheets 72K/11, 72K/10, 72K/6, 72K/7, 72K/14, 72K/15 (2010) with Scale 1:50,000 were initially georeferenced. Following georeferencing, the SOI toposheets and additional

maps are digitally enhanced with various elements, including points, lines, and polygons representing administrative boundaries, the network of roads and railroads, contours, drainage systems, and built-up areas. Next, topographic sheets were mosaicked so that the final image would provide a comprehensive perspective of the study area. The satellite photos were superimposed with these mosaicked images.

Image Classification

The USGS picture was georeferenced about the toposheets of the Survey of India. Initially, satellite data was analyzed using standard image processing methods such as band extraction, restoration, rectification, enhancement, and classification. This study employed Supervised Maximum Likelihood classification, the most prevalent remote sensing image data processing approach³⁴. Five main classes were used to classify land use and cover: wetland, built-up (residential and commercial), Barren land, Cropland, and Water body. Approximately ten training samples were employed in each class's Supervised Maximum Likelihood classification method²⁶. The following procedures were used in the analysis: (a) utilizing a polygon tool to digitize different classes, (b) showing all classes in a single layer, and (c) calculating the area of each class. Building a land use table comprising regions belonging to different classes allowed for identifying changes in the land use pattern following the analysis of the three years' worth of data²². We used maximum Supervised Maximum Likelihood classification to classify our LULC map into 5 classes such as – Agriculture, Forest, Waterbody, Barrenland, Built-up area.

Accuracy assessment

If the classification data is to help detect changes, accuracy evaluation is crucial for individual classifications²⁷. Accuracy evaluation aims to quantitatively examine how effectively the pixels were sampled into the proper land cover categories. Also, places discernible on the high-resolution Landsat image, Google Earth, and Map were the main emphasis when selecting pixels for the accuracy evaluation³⁸. By evaluating each pixel's classification against the limited land cover conditions derived from the matching ground truth data, this approach generated the overall correctness of the classified image³⁶. An error matrix and kappa coefficient were used to assess accuracy validation.

Since the kappa test considers all the items in the confusion matrix, not only diagonal elements, it was also used to gauge the degree of classification accuracy³⁵. In this study, classification accuracy was assessed using the Kappa coefficient derived from ground truth points collected through field surveys. Accuracy assessment was performed using ERDAS IMAGINE software. A total of 195 ground truth points representing different land-cover classes were used for the 2015 classification, while 178 and 201 ground truth points were used for the 2020 and 2025 classifications, respectively.

LULC map creation and change detection.

LU/LC map has been prepared for 3 years, namely 2015, 2020, and 2025 for three classes namely pre flood, during flood and post flood. The change in each class has been noted and analyzed. The study used the post-classification change detection technique performed in Arc-GIS Software. Because post-categorization is good at identifying the location, type, and pace of change in an urban setting, several studies have utilized it in this context¹⁴. The overlay approach was another method used to determine the changes in land cover/use throughout the designated period, and the resulting two-way cross-matrix was used to characterize the main changes in the research area. To ascertain the quantitative comparisons between one land cover type and another, as well as the equivalent area for each, during the assessed time, pixel by pixel, cross-tabulation analysis was performed. As a result, from the two five-class maps, a new thematic layer was created that included various combinations of "from-to" transition classes¹⁵. LULC maps were prepared for the years 2015, 2020, and 2025 using supervised classification with the Random Forest algorithm. Future land use/land cover for 2035 was predicted using the MOLUSCE (Modules for Land Use Change Simulation) model in QGIS. The MOLUSCE model simulates land use change by integrating transition probabilities, spatial drivers, and machine learning-based modelling to project future landscape dynamics. Inter-annual changes were analysed between successive years, and long-term trends were also assessed.

Future Post flood LULC Prediction Using CA-Markov Model

The CA-Markov model is regarded as the most dependable method and is employed by numerous

researchers to project land use and land cover (LULC) changes across both temporal and spatial dimensions⁴. The Markov model is a stochastic model that estimates change probabilities between one class and another, considering LULC changes over time¹². In other words, Markov chains are simply random values whose probabilities change depending on the previous number value²³.

The study employed a CA–Markov model to predict post-flood land use/land cover changes for the year 2035. The MOLUSCE model implemented in QGIS is classified as a Cellular Automata–Markov (CA–Markov) model that integrates Markov chain–based transition probabilities with cellular automata–driven spatial allocation to simulate future LULC dynamics. LULC maps for 2015, 2020, and 2025 were used as training inputs for model calibration. In addition, supplementary spatial variables, including Digital Elevation Model (DEM), elevation, slope, distance to roads, and distance to settlements, were incorporated as driving factors to enhance the prediction accuracy for 2035.

Result and Discussion

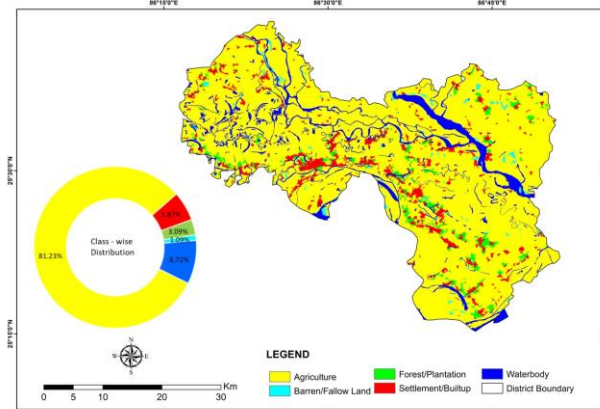


Figure 3. LULC 2015 before flood

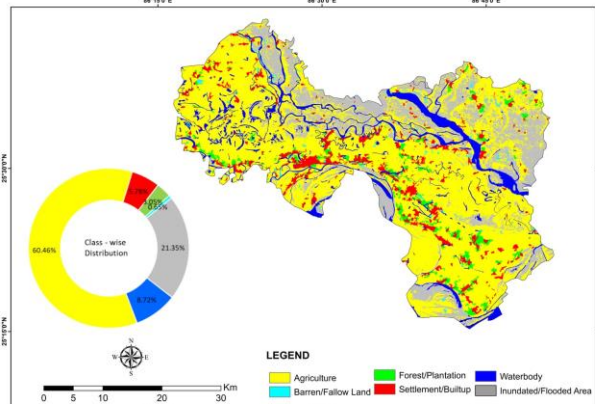


Figure 4. LULC 2015 during flood

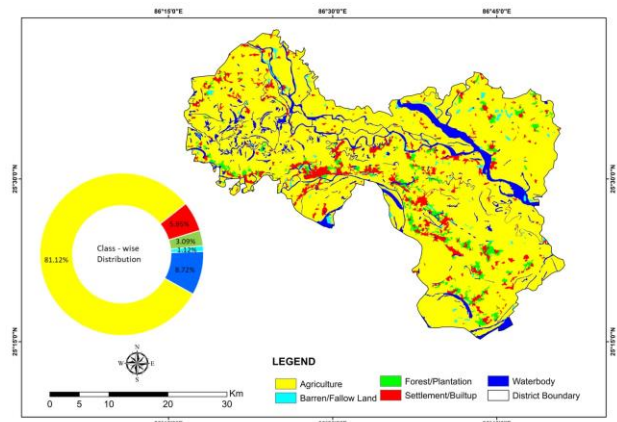


Figure 5. LULC 2015 after flood

Table 2. LULC and Area percentage (2015)

LULC 2015	2015 Before (ha)	Area %	2015 Flood During (ha)	Area %	2015 After (ha)	Area %
Waterbody	12955.48	8.72	12955.48	8.72	12955.48	8.72
Agriculture	120685.04	81.23	89826.63	60.46	120521.61	81.12
Settlement/Builtup	8721.18	5.87	8587.46	5.78	8840.03	5.95
Forest/Plantation	4590.87	3.09	4531.45	3.05	4590.87	3.09
Barren/Fallow Land	1619.43	1.09	950.86	0.65	1664.01	1.12
Inundated/Flooded Area			31720.12	21.35		
Total	148572	100	148572	100	148572	100

The LULC interpretation of 2015 has been divided into 3 categories pre flood LULC, during flood LULC and post-flood LULC as shown in figure 3, 4,5 and table 2: with 81.23% of the total area, agriculture is the most common activity in the pre-flood LULC. Waterbodies come in second with 8.72%, settlements with 5.87%, forests with 3.09%, and barrenland with 1.09%. This demonstrates that the region's economy is centered on agriculture. It has been noted that during floods, a significant amount of agricultural land has been submerged under floodwaters, decreasing the area from 81.23% prior to the flood season to 60.46%. This demonstrates that the region experiences agricultural losses during flood season. The emergence of a new category, Inundated/Flooded Area, which covers 31,720.12 hectares (21.35%), emphasizes the scope and intensity of flooding. Additionally, the bare land

exhibits a minor decline of 0.44 percent as a result of submergence. It is evident that the primary effect of the flood was on bare ground, which grew by 0.03, or 44.58 hectares, as a result of the flood, which also deposited silt and sand.

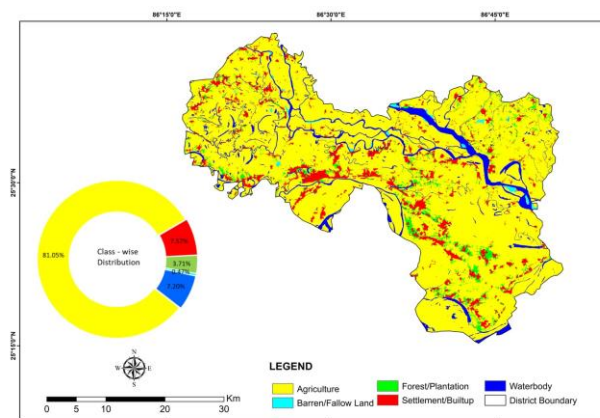


Figure 6. LULC 2020 before flood

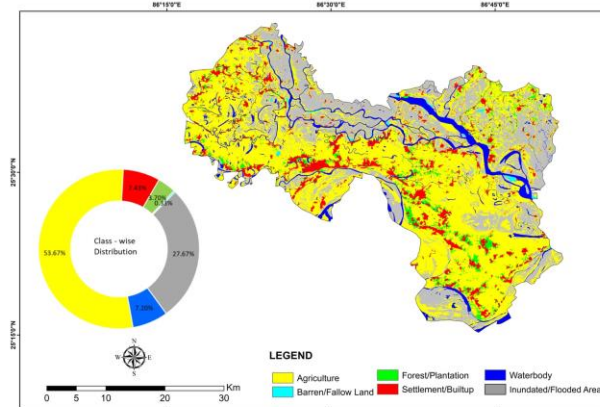


Figure 7. LULC 2020 during flood

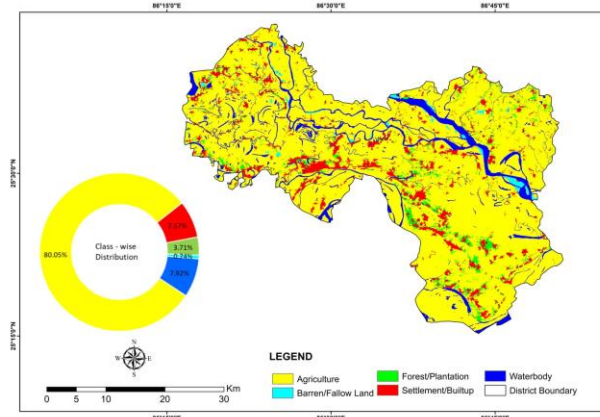


Figure 8. LULC 2020 after flood

LULC 2020	2020 Before (ha)	Area %	2020 Flood_During (ha)	Area %	2020 After (ha)	Area %
Waterbody	10697.18	7.20	10682.33	7.20	11752.05	7.92
Agriculture	120417.60	81.05	79738.59	53.67	118946.74	80.05
Settlement/Builtup	11246.90	7.57	11038.90	7.43	11261.76	7.57
Forest/Plantation	5512.02	3.71	5512.02	3.7	5512.02	3.71
Barren/Fallow Land	698.28	0.47	490.29	0.33	1099.43	0.74
Inundated/Flooded Area			41109.87	27.67		
Total	148572		148572		148572	

The LULC interpretation of 2020 has been divided into 3 categories pre flood LULC, during flood LULC and post-flood LULC figure 6, 7, 8 and table 3: With 81.05% of the total area, agriculture is the most common activity in the pre-food LULC. Waterbodies come in second with 7.20%, settlements with 7.57%, forests with 3.71%, and barrenland with 0.47 percent. This demonstrates that the region's economy is centered on agriculture. A significant amount of agricultural land was submerged by floodwater during the flood, which reduced the area to 53.67% from 81.05% before to the flood season. This demonstrates that the region experiences agricultural losses during flood season. The intensity and spatial extent of floods are highlighted by the appearance of a new category, Inundated/Flooded Area, which occupies 41109.8724 ha (27.67%). Additionally, the bare land exhibits a minor decline of 0.27 percent as a result of submersion. Following the flood, it was found that the bare land was the area most affected, growing by 401.144 hectares as a result of the flood's deposit of silt and sand.

Table 3. LULC and Area percentage (2020)

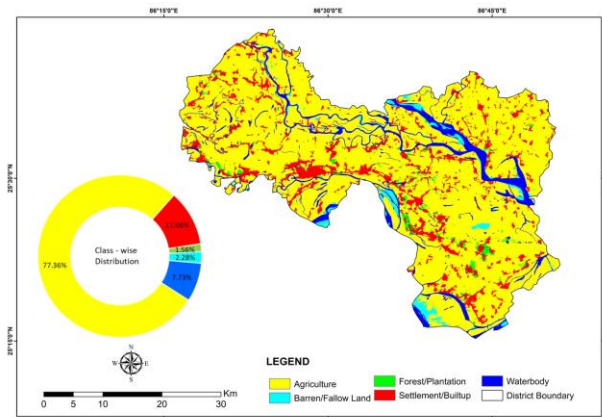


Figure 9. LULC 2025 Before flood

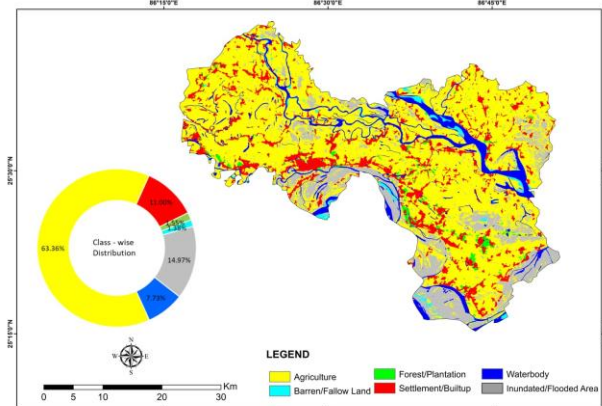


Figure 10. LULC 2025 during flood

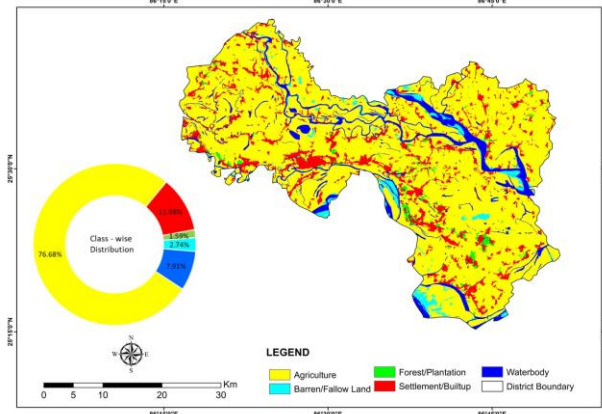


Figure 11. LULC 2025 after flood

Agriculture	11493 5.30	77.36	94135 .22	63.36	11392 5.01	76.68
Settlement/ Builtup	16446. 92	11.06	16342 .92	11	16461. 78	11.08
Forest/Plant ation	2317.7 2	1.56	2302. 87	1.55	2362.2 9	1.59
Barren/Fallo w Land	3387.4 4	2.28	2050. 29	1.38	4070.8 7	2.47
Inundated/Fl ooded Area			22256 .09	14.97		
Total	14857 2		14857 2		14857 2	

The LULC interpretation of 2025 has been divided into 3 categories pre flood LULC, during flood LULC and post-flood LULC figure 9, 10, 11: Agriculture accounts for 77.38% of the pre-food LULC, with waterbodies coming in second at 7.73%, settlements at 11%, forests at 1.56%, and barrenland at 2.28%. This demonstrates that the region's economy is centered on agriculture. A significant amount of agricultural land was submerged in floodwater during the flood, which reduced the area to 63.36% from 77.38% before to the flood season. This demonstrates that the region experiences agricultural losses during flood season. The emergence of a new category, Inundated/Flooded Area, which covers 22256.0856 ha (27.67%), emphasizes the scope and intensity of flooding. Additionally, the bare land exhibits a minor decline of 0.27 percent as a result of submersion. There is a noticeable rebound of agricultural land following the flood. At 113,925.01 ha (76.68%), agriculture nearly reaches its pre-flood level. This shows that as floodwaters subside, the majority of the flooded crops is once again usable. Similarly, the percentage of barren or fallow land rises to 2.47%, which could indicate recently exposed or temporarily damaged land surfaces following flooding and sediment deposition. Waterbodies exhibit a slight increase to 7.91%, which may be caused by leftover water in canals and low-lying places.

Table 4. LULC and Area percentage (2025)

LULC 2025	2025 Befe	Area %	2025 Duri	Area %	2025 After	Area %
Waterbody	11484. 62	7.73	11484 .62	7.73	11752. 05	7.91

Future Prediction:

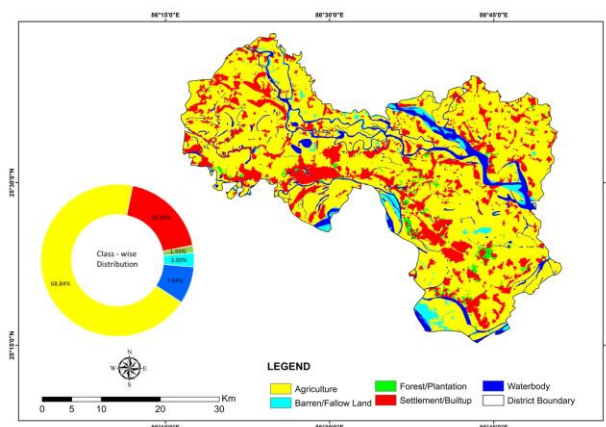


Figure 11. Future LULC Khagaria district, 2035

Table 5. LULC Future Prediction and Area Percentage for 203

LULC_2035	2035 Predicted	%
Waterbody	11799.35	7.94%
Agriculture	102283.47	68.84%
Settlement/Builtup	27728.84	18.66%
Forest/Plantation	2298.18	1.55%
Barren/Fallow Land	4462.13	3.00%
Total	148572	

The 2035 LULC map as shown in figure 11. was created using the CA–Markov model, which was based on the lulc maps from 2015, 2020, and 2025 and predicted how lulc will change in that year. According to the LULC table for 2035, the area under water would be 7.94%, agriculture would be 68.84%, settlement would be 18.66%, forests would be 1.55%, and barren land would be 3%. When comparing the LULC structure between 2025 (the baseline) and 2035 (the projected year), it is evident that the land-use pattern has changed, primarily due to the rapid growth of cities and the loss of agricultural land.

Table 5. shows the interpretation of LULC 2025, the agriculture sector shows the biggest shift, with a loss of roughly 12,651.82 ha (–8.52%) from 114,935.30 ha (77.36%) in 2025 to 102,283.48 ha (68.84%) in 2035. Strong pressure on agricultural land over the next ten years is indicated by this decline. A net gain of 11,281.93 ha (+7.60%) is demonstrated by the stark increase in settlement/built-up area from 16,446.92 ha (11.06%) to 27,728.85 ha (18.66%),

highlighting the rapid growth of infrastructure and urbanization.

From 11,484.62 ha (7.73%) in 2025 to 11,799.36 ha (7.94%) in 2035, the waterbody area increased somewhat, indicating a small expansion or enhancement of surface water features. The amount of barren/fallow land also increases from 3,387.44 ha (2.28%) to 4,462.13 ha (3.00%), suggesting that there may be transitional land uses or land degradation.

The amount of forests and plantations, however, has barely decreased from 1.56% to 1.55%. Overall, the comparison shows that while natural land covers mostly stay the same, the future landscape will be dominated by growing built-up regions at the expense of agricultural land.

Accuracy Assessment

The accuracy of the LULC maps computed in Eradas Imagine software is shown in table 6. The classification accuracy was determined to be 95.24 percent for Landsat-8 (2015) before flood imagery, 90.91 percent during flood, and 91.30 percent after flood; 91.38% for Sentinel-2 (2020) before flood imagery, 92.19 percent during flood, and 92.19% after flood; and 95.00% for Sentinel-2 (2025) before flood imagery, 95.95% during flood, and 95.24% after flood. The dependability of the results is measured by the Kappa coefficient, which was 93.99%, 89.01%, and 88.95% for 2015, 89.13%, 90.51%, and 95.45% for 2015, and 93.70%, 95.08%, and 90.57% for 2025, respectively.

Table 6. Accuracy Assessment of LU/LC Maps

Sr. No.	LULC	Overall Accuracy	Kappa Accuracy
1	2015 Before	95.24%	93.99%
2	2015 During	90.91%	89.01%
3	2015 After	91.30%	88.95%
4	2020 Before	91.38%	89.13%
5	2020 During	92.19%	90.51%
6	2020 After	96.43%	95.45%
7	2025 Before	95.00%	93.70%
8	2025 During	95.95%	95.08%
9	2025 After	92.54%	90.57%

Table 7. LULC 2015 and 2025 Change Map

Land Cover Category	2015 Area (ha)	2025 Area (ha)	Change (ha)	2015%	2025 %	Share Change (%)	Relative Change (%)	Trend
Waterbody	12,955.48	11,484.62	-1,470.86	8.72	7.73	-0.99	-11.36	Decrease
Agriculture	1,20,685.04	1,14,935.30	-5,749.74	81.23	77.36	-3.87	-4.76	Decrease
Settlement / Built-up	8,721.18	16,446.92	7,725.74	5.87	11.06	5.19	88.57	Increase
Forest / Plantation	4,590.87	2,317.72	-2,273.15	3.09	1.56	-1.53	-49.52	Decrease
Barren / Fallow Land	1,619.43	3,387.44	1,768.01	1.09	2.28	1.19	109.19	Increase

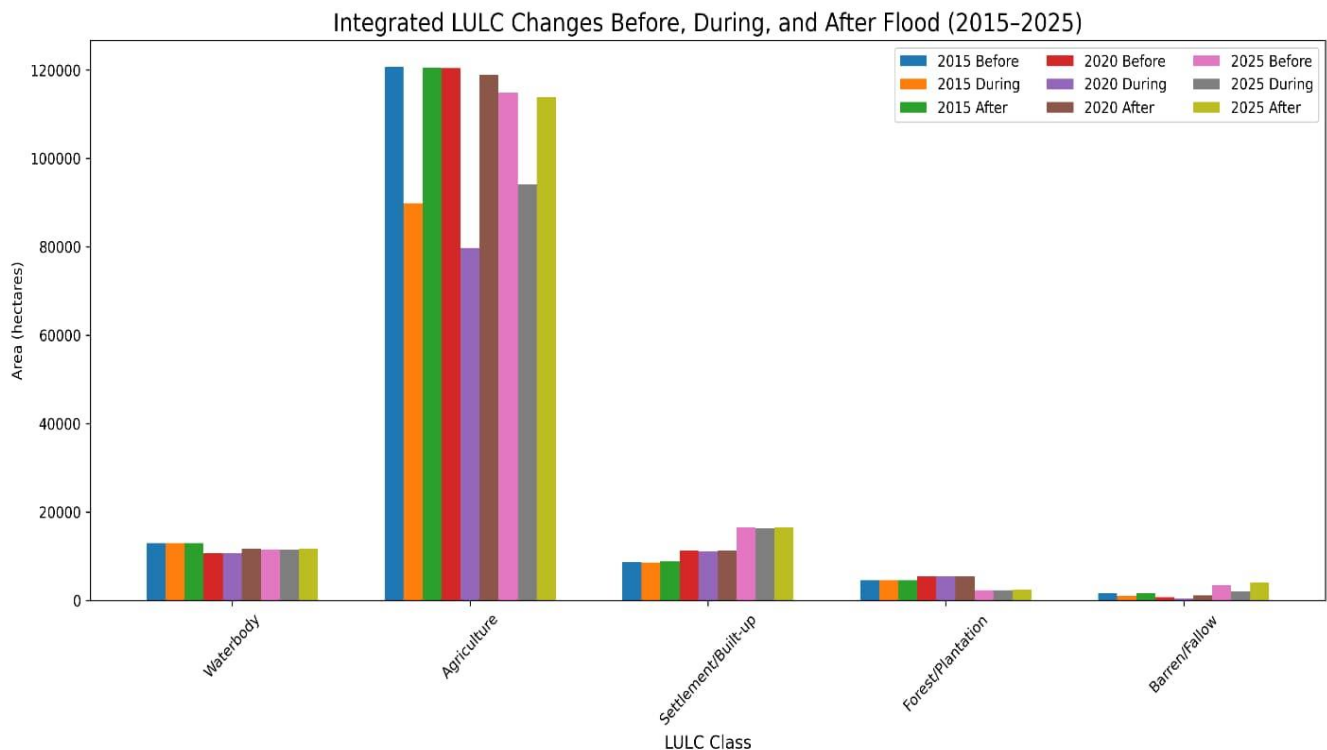


Figure 12. Integrated LULC Changes Before, During and After Flood (2015-2025)

The entire change over a ten-year period, from 2015 to 2025, is displayed in Table 7. and figure 12, Waterbodies fell by 0.99%, agriculture fell by 3.86%, settlements climbed by 5.19%, forests fell by 1.53%, and barrenland rose by 1.19%, according to the trend. This indicates that the flood had an effect on agriculture since it reduced land area while increasing barren land, possibly as a result of silt and

sand deposits brought on by flood inundation and hence also increasing the barrenland area by 1.19%.

Conclusion

The paper shows LULC change analysis that has been done for 3 decades in the Khagaria district of Bihar to determine the LU/LC transformation. This study examined how land use and land cover transformed over 20 years and how they changed. Using remotely sensed data such as Landsat 8 (2015),

Sentinel-2 (2020) and Sentinel-2 (2025), spatiotemporal analysis of land use and land cover is conducted from 2005 to 2025. The study generated five classes—built-up farmland, barren terrain, water bodies, and wetlands using a random forest classifier to perform supervised classification on the LU/LC map. Overall Accuracy for the 2015 LU/LC map was 95.24%; and for the 2025 LU/LC map, it was 95.0%. It was found that the major change in LU/LC occurred due to flood. Flood is a natural occurrence in this district. Floods had the most impact on the agricultural sector, which, despite still being the most agriculturally dominated sector, had a 3.87% decline following the flood occurrence. The built-up area has increased by 5.19%, with part located close to flood-prone areas and others farther away. Waterbody has decreased by 0.99%. Barren Land/Fallow Land have increased by 1.19% and Forest/Plantation area decreased by 1.53% because of floods. To make up for losses caused by natural catastrophes, the government should enact effective laws, such as those that mandate insurance on assets that are considered high-risk. Finding the main reasons behind adverse land cover change would be easier by using LU/LC.

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